

Table 1: Sequential MAS results for DeepSeek-R1-Distill-Llama-8B. Best results in each row are bold. Kernel and Kernel-ES are highlighted.

Tasks	Metrics	Baselines		LatentMAS variants					Improve
		Single	TextMAS	Identical	Linear	Kernel	Kernel-ES	Soft	
Sequential MAS Setting									
AIME24	Acc.	45.00	34.58	12.50	0.00	13.33	12.92	12.92	↓ 21.3
	Token	10809	12138	12249	20000	11642	11992	33557	↓ 4.1%
	Time	217.3	377.8	296.7	330.6	288.4	303.3	835.4	× 1.3
AIME25	Acc.	35.42	27.08	10.83	12.08	10.00	8.33	8.33 [†]	↓ 17.1
	Token	11844	12822	12719	12727	12767	12923	32396 [†]	↓ 0.4%
	Time	243.9	398.1	300.7	302.0	297.3	334.0	850.0 [†]	× 1.3

Note. Values are computed from `result_ds`. Acc. is accuracy (%); Token is the mean number of generated text tokens per problem; Time is the mean elapsed time per sample in seconds. Unless marked with [†], entries are averages over eight repetitions (seeds 42–49). [†] denotes a currently incomplete aggregate: Identical–Hierarchical on AIME25 uses three repetitions and Soft–Sequential on AIME25 uses two. Improve compares the better of Kernel and Kernel-ES with TextMAS; missing experiment combinations are denoted by –.

Table 2: Hierarchical MAS results for DeepSeek-R1-Distill-Llama-8B. Best results in each row are bold. Kernel and Kernel-ES are highlighted.

Tasks	Metrics	Baselines		LatentMAS variants					Improve
		Single	TextMAS	Identical	Linear	Kernel	Kernel-ES	Soft	
Hierarchical MAS Setting									
AIME24	Acc.	47.08	45.00	15.00	0.00	13.75	15.42	15.00	↓ 29.6
	Token	11114	30999	12662	20000	12422	12271	33308	↓ 60.4%
	Time	228.4	715.6	299.3	331.6	302.2	310.9	871.1	× 2.4
AIME25	Acc.	31.67	32.92	8.89 [†]	—	—	—	—	—
	Token	11936	33376	14145 [†]	—	—	—	—	—
	Time	250.3	799.4	321.8 [†]	—	—	—	—	—

Note. Values are computed from `result_ds`. Acc. is accuracy (%); Token is the mean number of generated text tokens per problem; Time is the mean elapsed time per sample in seconds. Unless marked with [†], entries are averages over eight repetitions (seeds 42–49). [†] denotes a currently incomplete aggregate: Identical–Hierarchical on AIME25 uses three repetitions and Soft–Sequential on AIME25 uses two. Improve compares the better of Kernel and Kernel-ES with TextMAS; missing experiment combinations are denoted by —.